

Class 5: ggplot2 viz

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Background

There are many graphics systems in R for making plots and figures. These include so-called “*base R*” *graphics* like `plot()` and **ggplot2** which is an add on package.

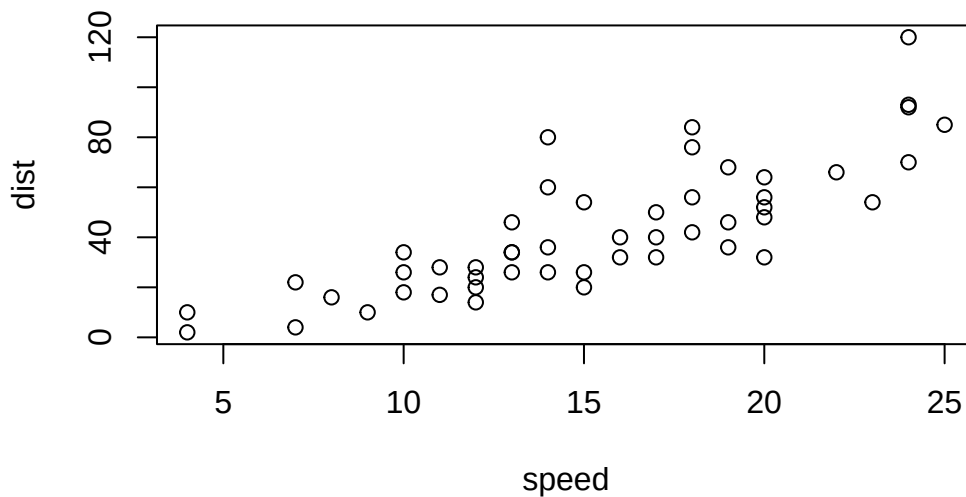
Let’s compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Before I can use ggplot2 I need to install it on my computer.

To install it, we can use the function `install.packages()`, however we never run this function in the document because it would redownload it constantly. Instead this is done in the console.

Below are the 3 commands used for the libraries:

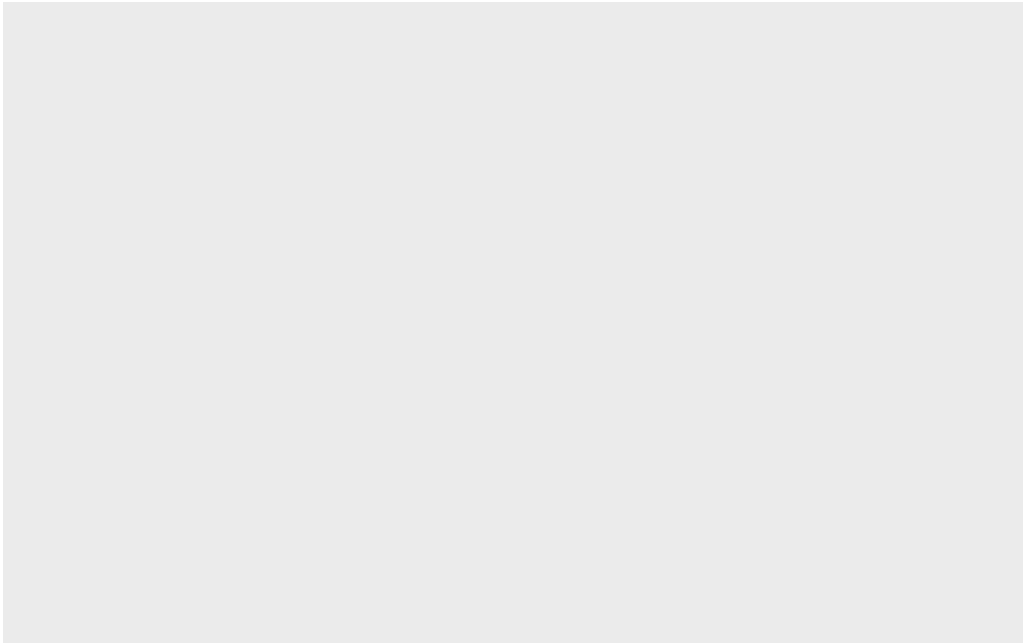
```
# install.packages("ggplot2")  
# install.packages("gapminder")  
# install.packages("dplyr")
```

Once installed, we need to load up the package into our R brain:

```
library(ggplot2)  
library(gapminder)  
library(dplyr)
```

The main function in ggplot2's package is called `ggplot()`

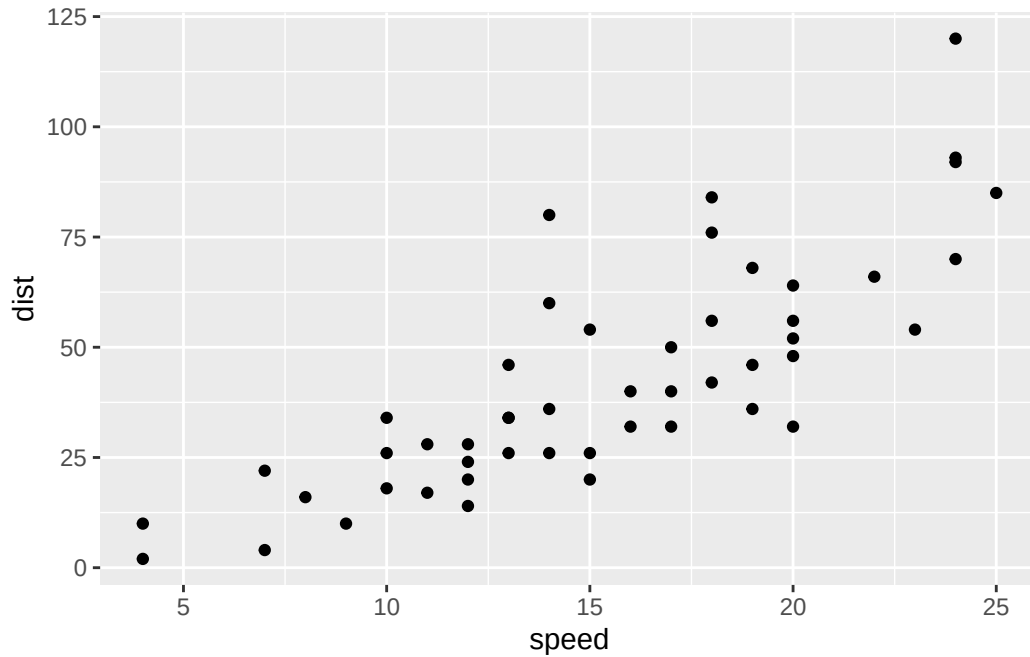
```
ggplot(cars)
```



Every ggplot has at least 3 layers:

- the **data** (data.frame of what we want to plot)
- the **aes** (how data maps onto plot)
- the **geom** (how we want the plot to be drawn)

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point()
```

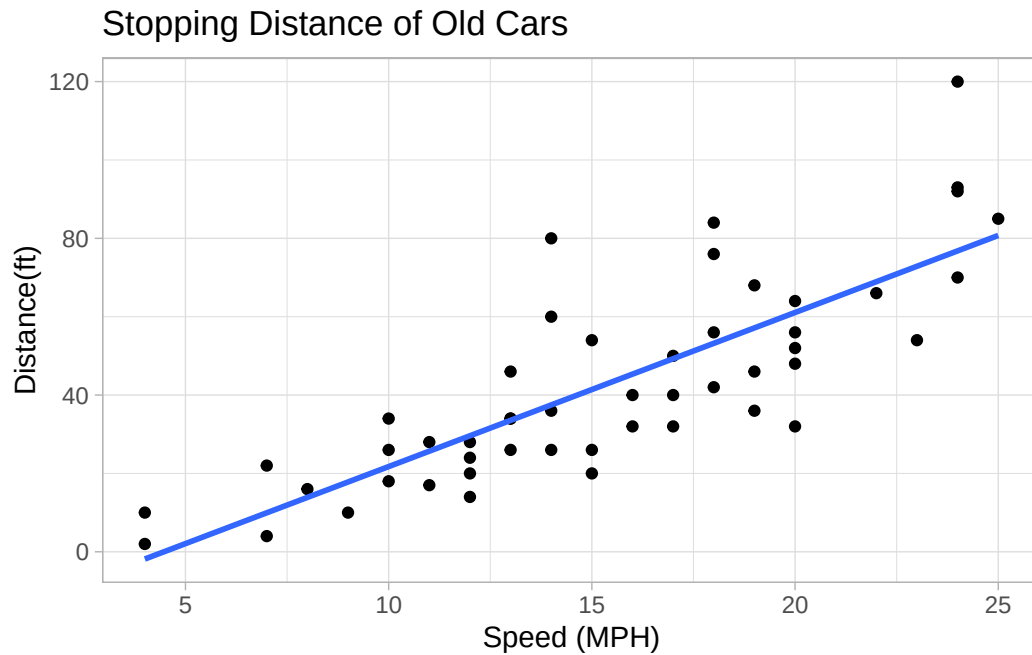


Add some customization/features

Let's add a trendline that shows the relationship between speed + distance.

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth(method="lm", se=FALSE) +  
  theme_light() +  
  labs(title="Stopping Distance of Old Cars", x="Speed (MPH)", y="Distance(ft)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Figure

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
head(genes[genes$State == "up",])
```

	Gene	Condition1	Condition2	State
10	ABCC3	0.9305738	3.260304	up
11	ABCC5	4.6004252	5.499443	up
420	AHNAK	6.6284895	7.611752	up

421	AHNAK2	3.6510510	5.181916	up
496	ALPPL2	0.5586050	2.625953	up
500	AMIGO2	3.1435673	4.817252	up

```
table(genes$State)
```

down	unchanging	up
72	4997	127

```
round( table(genes$State)/nrow(genes) * 100, 2 )
```

down	unchanging	up
1.39	96.17	2.44

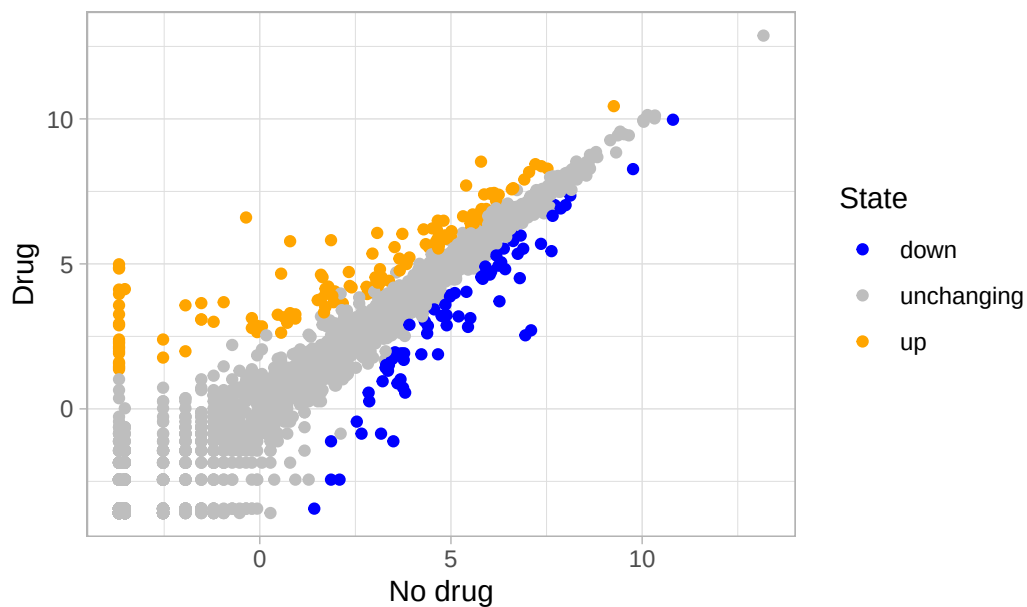
```
colnames(genes)
```

```
[1] "Gene"          "Condition1" "Condition2" "State"
```

```
p <- ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  theme_light() +
  labs(x="No drug", y="Drug", title="Expression chnages upon GLP-1 inhibitor")

p + scale_color_manual( values=c("blue","gray","orange"))
```

Expression changes upon GLP-1 inhibitor



Going further into the GapMinder dataset

Here is the read of it

```
head(gapminder)
```

```
# A tibble: 6 x 6
  country    continent  year lifeExp      pop gdpPercap
  <fct>      <fct>    <int>  <dbl>    <int>    <dbl>
1 Afghanistan Asia      1952   28.8  8425333    779.
2 Afghanistan Asia      1957   30.3  9240934    821.
3 Afghanistan Asia      1962   32.0 10267083    853.
4 Afghanistan Asia      1967   34.0 11537966    836.
5 Afghanistan Asia      1972   36.1 13079460    740.
6 Afghanistan Asia      1977   38.4 14880372    786.
```

Q. How many entries are there?

```
nrow(gapminder)
```

```
[1] 1704
```

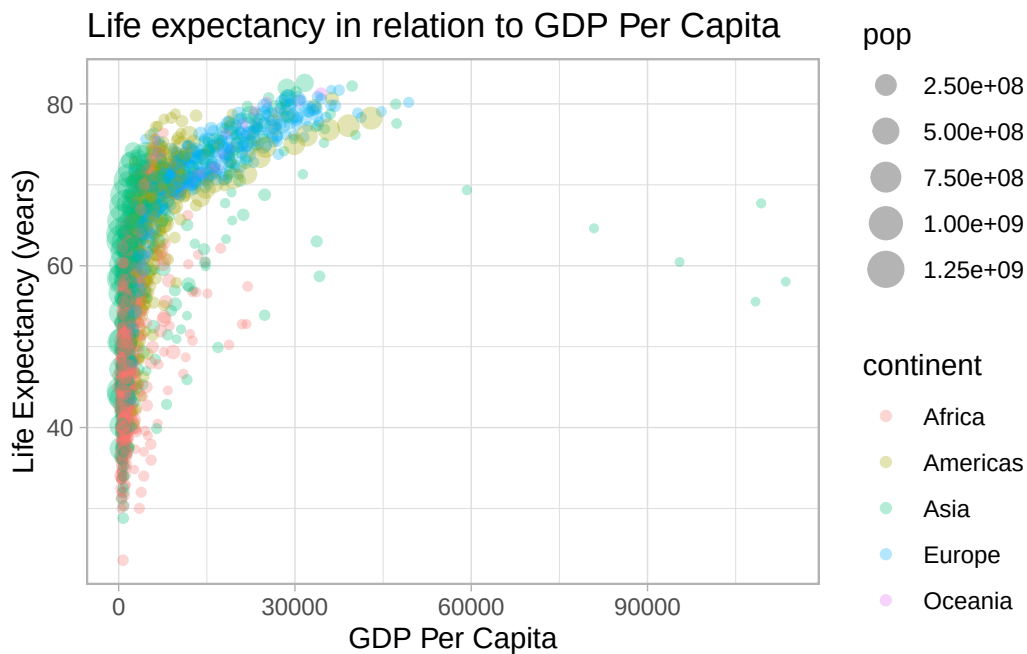
Q. How many countries are in this dataset?

```
length(unique(gapminder$country))
```

```
[1] 142
```

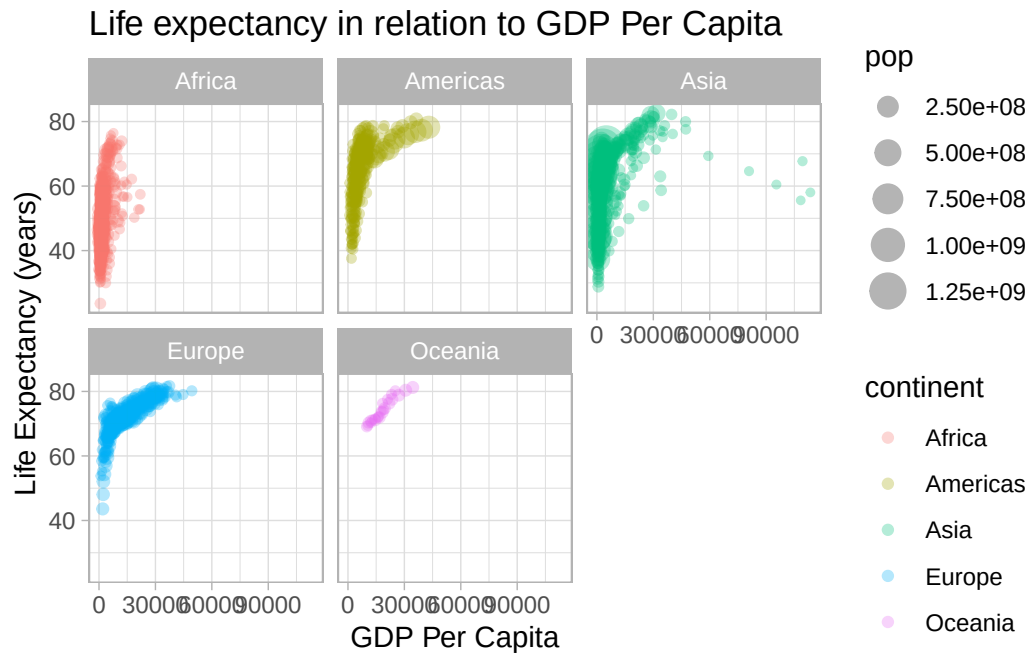
Let's make our first plot of this gapminder dataset

```
p <- ggplot(gapminder) +  
  aes(gdpPerCap, lifeExp, col=continent, size=pop) +  
  geom_point(alpha=0.3) +  
  theme_light() +  
  labs(x="GDP Per Capita", y="Life Expectancy (years)", title="Life expectancy in relation to  
p
```



I can add more layers to change the call of p

```
p +  
  facet_wrap(~gapminder$continent)
```

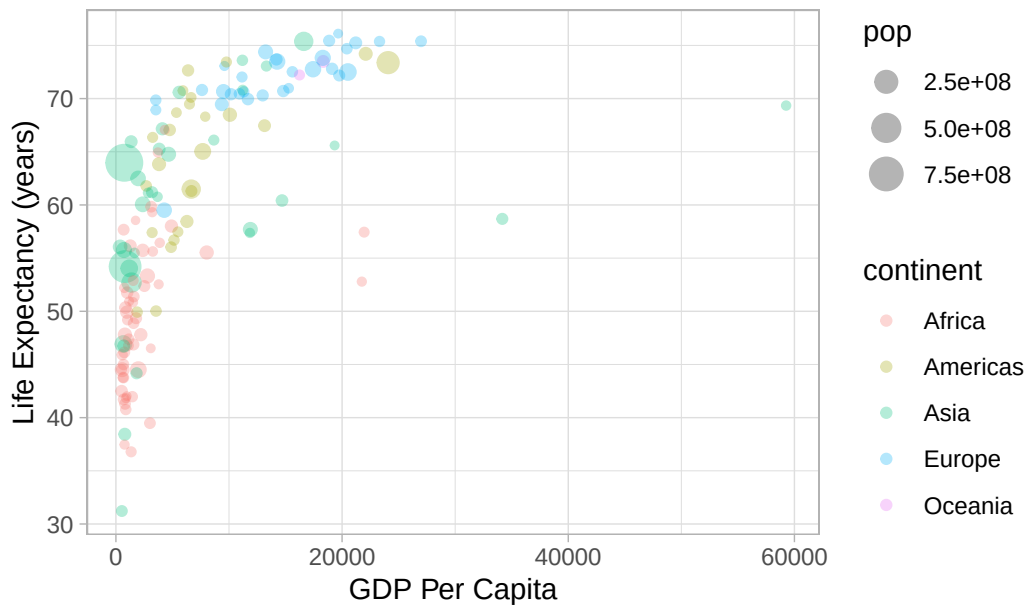
Make a plot for 1977 and 2007

```
gapminder_1977 <- p +
  gapminder %>% filter(year==1977) +
  labs(title="Life expectancy in relation to GDP Per Capita, 1977")

gapminder_2007 <- p +
  gapminder %>% filter(year==2007) +
  labs(title="Life expectancy in relation to GDP Per Capita, 2007")

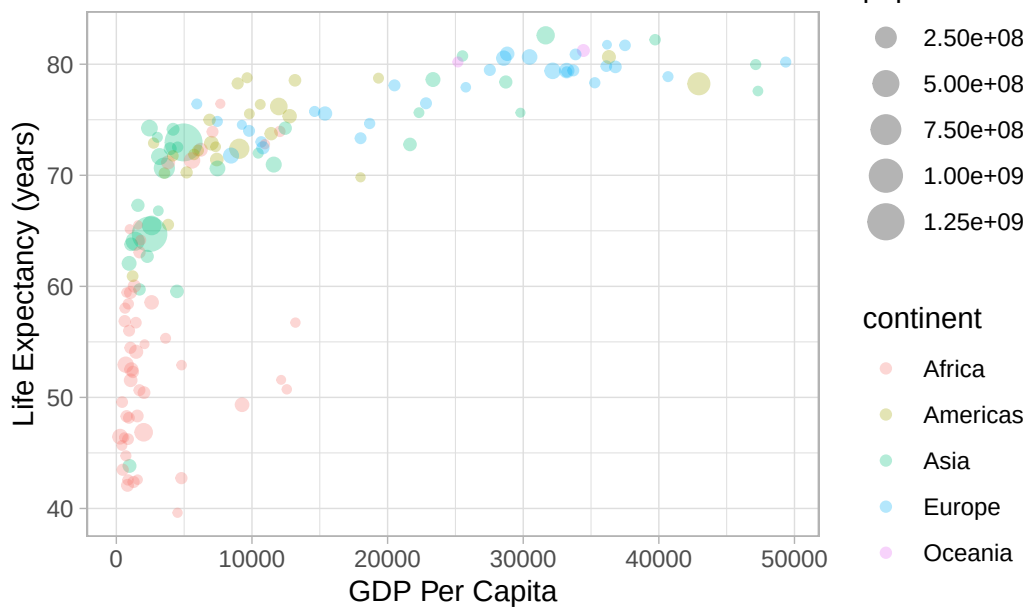
gapminder_1977
```

Life expectancy in relation to GDP Per Capita, 1997



gapminder_2007

Life expectancy in relation to GDP Per Capita, 2007



Q. Make a histogram of lifeExp colored by continent (try using `fill=continent` or `col=continent`)

```
ggplot(gapminder) +  
  aes(lifeExp, fill=continent) +  
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

